Code No: 136CX **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February - 2023
MOBILE COMPUTING
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	What is meant by mobile computing?	[2]
b)	What are the tele services of GSM Protocol?	[3]
c)	Write about the exposed terminal problem?	[2]
d)	What is meant by encapsulation in Mobile IP Protocol?	[3]
e)	What is Capture effect in TCP over ad hoc links?	[2]
f)	State the advantage of Snooping TCP protocol.	[3]
g)	Briefly write about Push model for data delivery to Mobile host?	[2]
h)	What is the role of indexing technique in mobile data dissemination?	[3]
i)	State the disadvantage of reactive protocols for Manet routing.	[2]
j)	What is meant by multi-hop communication?	[3]

PART – B

(50 Marks)

2.a) Describe briefly about GSM protocol.
b) Explain the difference between MOC and MTC in GSM. [5+5]

OR

3. With a neat sketch, write about the distinct components of GPRS compared to GSM. [10]

4. Explain how the problems of IP for host mobility are addressed using Mobile IP. [10]

OR

5. Write about IEEE 802.11 standard giving its protocol architecture. [10]

6.a) Illustrate the effect of TCP congestion control policy on mobile links.
b) Explain how S-TCP overcomes the disadvantages of I-TCP. [5+5]

OR

7.a) Explain the terms- hoarding, caching, and Adaptation in mobile environment.
b) Explain the Kangaroo transaction model for mobile clients. [5+5]

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8. Illustrate the working mechanism of Broadcast Disk model with an example. [10]

OR

9.a) What is meant by selective tuning? How it helps in data delivery to mobile hosts.
b) Write about client-side adaptation in mobile environment. [5+5]

10. Explain how route discovery is done in reactive protocol for Manet routing. [10]

OR

11. Illustrate the Manet routing mechanism using DSDV protocol. [10]

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